

DATA STRUCTURES & ALGORITHMS FINAL EXAM

1. What is the worst-case time complexity of QuickSort?

- A. $O(n)$
- B. $O(n \log n)$
- C. $O(n^2)$
- D. $O(\log n)$

2. Which traversal of a binary search tree yields keys in sorted order?

- A. Preorder
- B. Inorder
- C. Postorder
- D. Level-order

3. Which graph algorithm finds shortest paths from a single source with non-negative edge weights?

- A. Prim
- B. Kruskal
- C. Dijkstra
- D. Bellman-Ford

4. Consider a hash table of size 10 using open addressing with linear probing. The hash function is $h(k) = k \bmod 10$.

What will be the final index assigned to key 42 after all collisions are resolved?

- A. Index 2
- B. Index 3
- C. Index 4
- D. Index 5

5. True or False: In a min-heap, the root node always contains the minimum element in the tree.

- A. True
- B. False